

Summary of camera-ready changes — D2AI 2026, DM2158

Mining Attention Dynamics: Auditing KV-Saliency Prediction in LLMs

We thank the reviewer for the four concerns below. The revised manuscript treats KVSaliencyBench as a measurement and benchmark contribution. This summary describes the revision; it does not claim new GPU experiments during the final editorial pass.

1. Attention masking versus physical eviction. The abstract, §VII, §IX and conclusion scope answer-quality findings to the evaluated masked loop. Logical retention is distinguished from physical storage and serving efficiency. The separate physical-KV reference reports storage and parity checks; extended mask/physical tests exceed the preset numerical tolerance. Neither scoring cost nor simulator measurements support a production TPOT or throughput claim.

2. Offline predictor versus runtime scorer. §IV-B and Table I specify the different feature realizations, and Exp#8 explicitly identifies its policy as the runtime reconstruction rather than the offline predictor of Table II. The bridge compares feature realizations on matched rows and separately along an on-policy trajectory. Changing trajectory and candidate set together does not identify the cause of the historical proxy gap. The draw.io overview shows the distinct offline, masked-loop and physical-reference paths.

3. Scope of query–key redundancy. §IX discloses the legacy query/key coordinate mismatch and withdraws the “faithful Quest” interpretation. Version-2 phase-aligned token/head dot-max controls are retained and have small, opposite-signed changes in the two tested model configurations. They are neither Quest’s page-bound algorithm nor evidence of intrinsic query redundancy. Exp#9 retains the synthetic needle experiment as a negative control.

4. External validity. §IX separates the main two-model evidence from historical workload, 14B and 16K extensions. Findings do not generalize to MoE, base models, contexts of at least 32K, or temperature sampling. Exp#8 reports mean equivalence within a specified tolerance on seven QA datasets, with task heterogeneity visible in the forest plot; both compressed policies lose quality relative to full cache.

The introduction and conclusion identify versioned collection, provenance and an explicit benchmark protocol as contributions to data and database systems for AI. The README and English presentation lead with the same v2 evidence as the manuscript. Historical results remain labeled and unchanged.